



Building a Home SOC Lab with Wazuh SIEM

DETECT • MONITOR • ANALYZE • RESPOND



Centralized Visibility

Aggregate and monitor logs from all assets



Threat Detection

Identify suspicious activity in real time



Incident Response

Investigate, triage and respond effectively



Security Insights

Turn data into actionable security intelligence



- Reconnaissance
- Exploitation Attempts
- Adversary Simulation



- Windows Events
- System Activity
- Application Logs



- Real-time Monitoring
- Threat Detection
- Alerting & Reporting



BUILD. LEARN. DEFEND.

Hands-on SOC experience in your home lab.



Skill Development
Practical cybersecurity skills for real-world scenarios.



Career Advancement
Stand out with in-demand SIEM and SOC expertise.



Stronger Defense
Proactive threat hunting and incident response capabilities.



Community Driven
Join a global community of defenders and contributors.

SOC ARCHITECTURE DIAGRAM

Attack Flow, Log Flow & Monitoring

→ **ATTACK FLOW**
Kali → Windows

---→ **LOG FLOW**
Windows → Wazuh SIEM

---→ **DASHBOARD ACCESS**
Analyst → Dashboard



SECURITY BENEFITS

- Real-time Threat Detection
- Centralized Log Management
- Incident Investigation & Response
- Visibility & Compliance

1 ATTACKER MACHINE



Kali Linux
(Attacker Machine)



ATTACK TRAFFIC

2 ENDPOINT



Windows 10 Workstation

Wazuh Agent Installed



SECURITY LOGS

3 WAZUH SIEM (LOG FLOW)



Wazuh Manager

- Log Collection
- Agent Management
- Rule Engine
- Alerting



Wazuh Indexer

- Log Indexing
- Data Storage
- Search & Analytics



Wazuh Dashboard

- Visualization
- Threat Monitoring
- Reports
- Compliance



DASHBOARD ACCESS



SOC Analyst
(Dashboard Access)

ARCHITECTURE OVERVIEW



ATTACK FLOW

Kali Linux (Attacker Machine) sends malicious traffic to the Windows 11 Workstation.



LOG FLOW

The Wazuh Agent on Windows collects security events and forwards logs to the Wazuh SIEM for processing.



WAZUH SIEM

Wazuh Manager collects and analyzes logs, Indexer stores and indexes data, and Dashboard visualizes insights.



DASHBOARD ACCESS

SOC Analyst accesses the Wazuh Dashboard to monitor alerts, investigate threats, and respond to incidents.



Centralized Logging



Threat Detection



Real-time Alerts



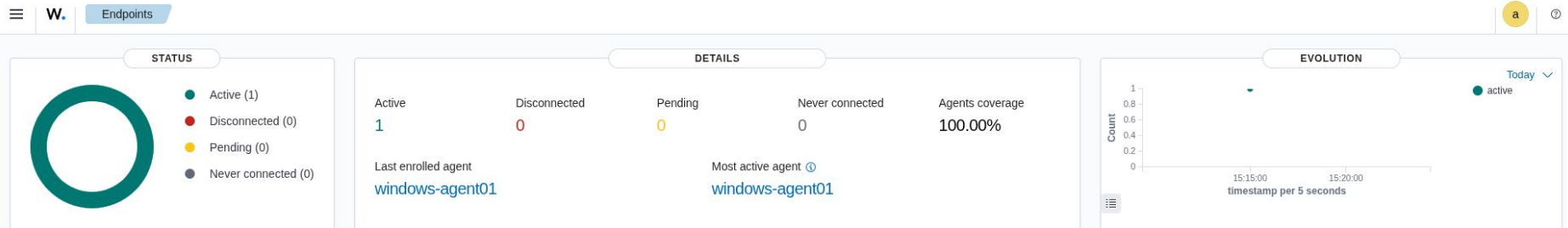
Visibility & Monitoring



Incident Response



Compliance Ready



Agents (1)

status=active

Deploy new agent

Refresh

Export formatted

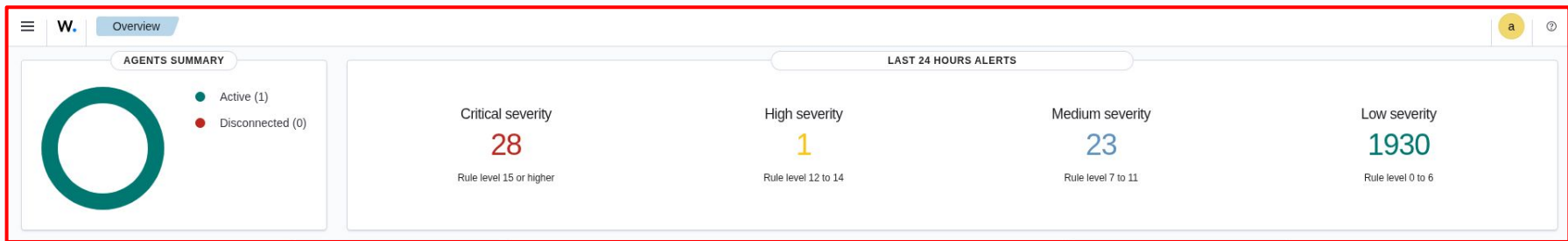
WQL

Refresh

ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
001	windows-agent01	10.0.2.15	default	Microsoft Windows 10 Pro 10.0.19045.6456	node01	v4.8.2	active	

Rows per page: 10

< 1 >



MachineView

Recycle Bin

Microsoft Edge

Administrator: Windows PowerShell

PS C:\Users\issame\Downloads> ls

Directory: C:\Users\issame\Downloads

Mode	LastWriteTime	Length	Name
d----	6/25/2026 9:53 AM		Sysmon
-a----	6/25/2026 9:52 AM	2909385	Sysmon.zip

PS C:\Users\issame\Downloads> cd .\Sysmon\
PS C:\Users\issame\Downloads\Sysmon> ls

Directory: C:\Users\issame\Downloads\Sysmon

Mode	LastWriteTime	Length	Name
-a----	6/25/2026 9:34 AM	7490	Eula.txt
-a----	6/25/2026 9:34 AM	6258072	Sysmon.exe
-a----	6/25/2026 9:34 AM	3250120	Sysmon64.exe
-a----	6/25/2026 9:34 AM	3141024	Sysmon64a.exe
-a----	6/25/2026 9:52 AM	123257	sysmonconfig-export.xml

PS C:\Users\issame\Downloads\Sysmon> .\Sysmon64.exe -c .\sysmonconfig-export.xml

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Configuration updated.

PS C:\Users\issame\Downloads\Sysmon>

Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS C:\Users\issame\Downloads\Sysmon> Get-Service Sysmon64

Status	Name	DisplayName
Running	Sysmon64	Sysmon64

PS C:\Users\issame\Downloads\Sysmon>

</localfile>

<localfile>
<location>System</location>
<log_format>eventchannel</log_format>
</localfile>

<localfile>
<location>active-response\active-responses.log</location>
<log_format>syslog</log_format>
</localfile>

<localfile>
<location>Microsoft-Windows-Sysmon/Operational</location>
<log_format>eventchannel</log_format>
</localfile>

<!-- Policy monitoring -->
<rootcheck>

Ln 61, Col 29100%Windows (CRLF)UTF-8

Packages (17)

[Refresh](#) [Export formatted](#)

Search				WQL
Name ↑	Architecture	Version	Vendor	
Microsoft 365 Copilot	x86_64	19.2606.55071.0	Microsoft Corporation	
Microsoft Edge	i686	149.0.4022.80	Microsoft Corporation	
Microsoft Edge WebView2 Runtime	i686	149.0.4022.80	Microsoft Corporation	
Microsoft OneDrive		26.106.0603.0003	Microsoft Corporation	
Microsoft Store	x86_64	22605.1401.13.0	Microsoft Corporation	
Microsoft Update Health Tools	x86_64	3.74.0.0	Microsoft Corporation	
OneNote for Windows 10	x86_64	16001.14326.22594.0	Microsoft Corporation	
Phone Link	x86_64	1.25072.79.0	Microsoft Corporation	
Solitaire & Casual Games	x86_64	4.25.4290.0	Microsoft Studios	
Tips	x86_64	10.2312.1.0	Microsoft Corporation	
Update for x64-based Windows Systems (KB5001716)	x86_64	8.94.0.0	Microsoft Corporation	
Wazuh Agent	i686	4.8.2	Wazuh, Inc.	
Xbox Game bar	x86_64	1.54.4001.0	Microsoft Corporation	
Xbox Identity Provider	x86_64	12.130.16001.0	Microsoft Corporation	
Rows per page: 25				< 1 >

Processes (119)

[Refresh](#) [Export formatted](#)

Search							WQL
Name ↑	PID	Parent PID	VM size	Priority	NLWP	Command	
AggregatorHost.exe	5080	2628	9625600	8	4	C:\Windows\System32\AggregatorHost.exe	
M365Copilot.exe	7680	4784	144293888	8	31	C:\Program Files\WindowsApps\Microsoft.MicrosoftOfficeHub_19.2606.55071.0_x-ww_8wekyb3d8bbwe\M365Copilot.exe	
MicrosoftEdgeUpdate.exe	2532	1404	7200768	8	5	C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe	
MoUsCoreWorker.exe	6164	844	58900480	8	10	C:\Windows\System32\MoUsCoreWorker.exe	
MusNotifyIcon.exe	8688	4960	6156288	6	4	C:\Windows\System32\MusNotifyIcon.exe	
OneDrive.Sync.Service.exe	5528	1424	103366656	8	15	C:\Users\issame\AppData\Local\Microsoft\OneDrive\26.106.0603.0003\OneDrive.Sync.Service.exe	
OneDrive.exe	10228	1424	154816512	8	20	C:\Users\issame\AppData\Local\Microsoft\OneDrive\OneDrive.exe	
RuntimeBroker.exe	3292	844	59342848	8	15	C:\Windows\System32\RuntimeBroker.exe	
RuntimeBroker.exe	7744	844	28659712	8	12	C:\Windows\System32\RuntimeBroker.exe	
RuntimeBroker.exe	2248	844	42278912	8	3	C:\Windows\System32\RuntimeBroker.exe	
Rows per page: 10							< 1 2 3 4 5 ... 12 >

data.win.system.eventID:4625

DQL

Today

Show dates

Refresh

manager.name: server2Vcpu4gb agent.id: 001 + Add filter

wazuh-alerts.*

Search field names

Filter by type 0

Selected fields

rule.description

rule.id

rule.level

Available fields

agent.id

agent.ip

agent.name

data.win.eventdata.authenticationPackageName

data.win.eventdata.failureReason

data.win.eventdata.ipAddress

data.win.eventdata.ipPort

data.win.eventdata.keyLength

data.win.eventdata.loginProcessName

data.win.eventdata.logonType

data.win.eventdata.processId

data.win.eventdata.processName

data.win.eventdata.status

data.win.eventdata.subjectDomainName

data.win.eventdata.subjectLogonId

data.win.eventdata.subjectUserName

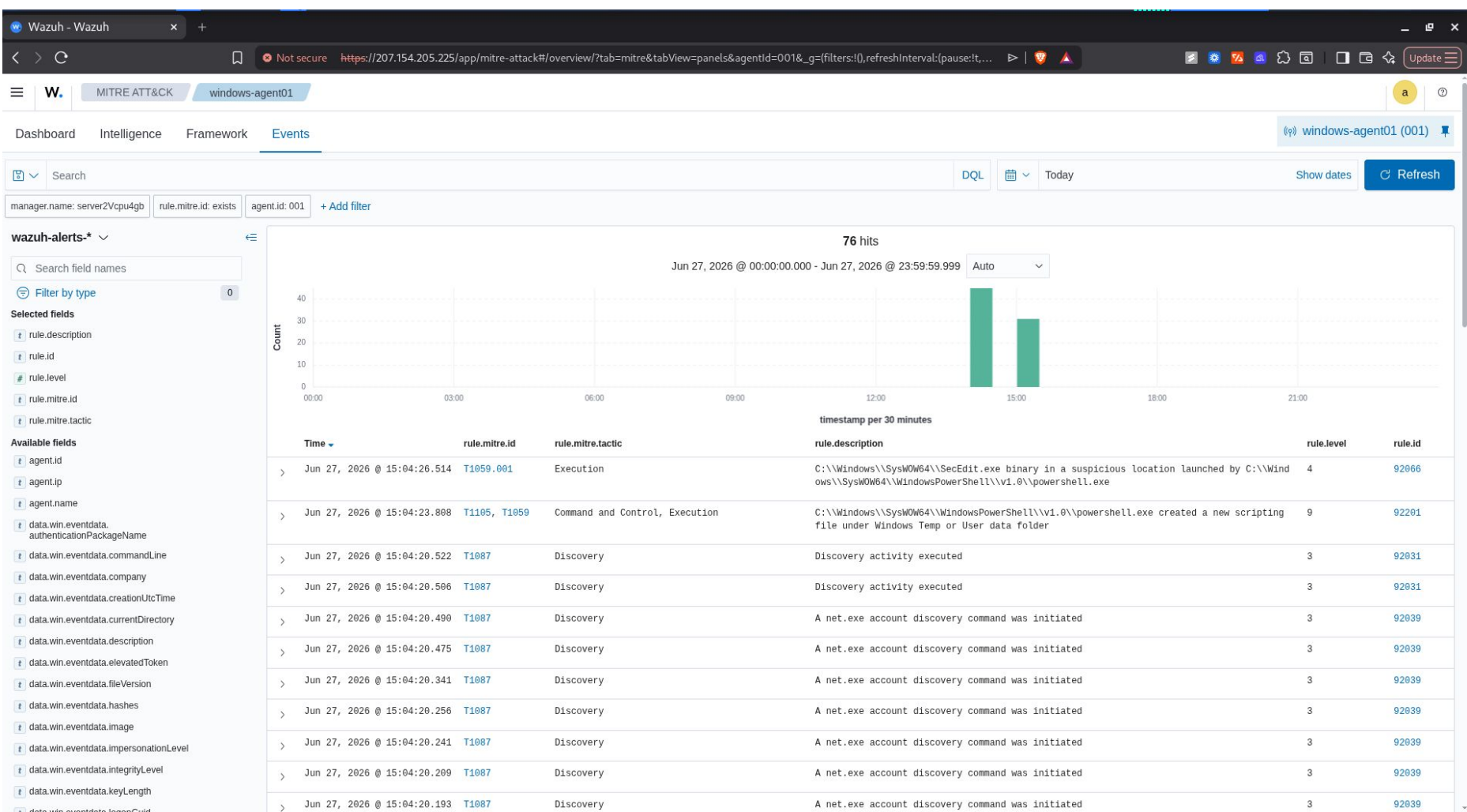
data.win.eventdata.subjectUserSid

data.win.eventdata.subStatus

data.win.eventdata.targetDomainName



Time	rule.description	rule.level	rule.id
> Jun 27, 2026 @ 15:44:41.481	Logon failure - Unknown user or bad password.	5	60122
> Jun 27, 2026 @ 15:44:37.990	Logon failure - Unknown user or bad password.	5	60122
> Jun 27, 2026 @ 15:44:35.182	Logon failure - Unknown user or bad password.	5	60122





CYBERSECURITY PROJECT TIMELINE

Building a Home SOC Lab with Wazuh SIEM



1

VPS DEPLOYMENT

Provisioned a secure Ubuntu VPS on DigitalOcean as the foundation for the Wazuh SIEM stack.



2

WAZUH INSTALLATION

Installed Wazuh Manager, Indexer, and Dashboard on the VPS and configured the environment.



3

WINDOWS ENDPOINT SETUP

Deployed Windows workstation and performed initial configuration for monitoring.



4

AGENT REGISTRATION

Installed Wazuh Agent on Windows and successfully registered the endpoint with the Manager.



5

VERSION COMPATIBILITY TROUBLESHOOTING

Identified and resolved version mismatches between Wazuh components to ensure proper communication.



6

DASHBOARD RECOVERY

Troubleshot dashboard service and API issues and restored full visibility in the Wazuh UI.



7

SUCCESSFUL LOG COLLECTION

Validated end-to-end log collection from Windows endpoint to Wazuh SIEM. Monitoring is now fully operational.

PROJECT OUTCOMES



Centralized Monitoring
All endpoint logs are centrally collected and visible in one place.



Threat Visibility
Real-time detection of suspicious activities and security events.



Alerting
Configured alerts for critical events and anomalies.



Operational SOC Lab
A fully functional SOC lab environment for learning and monitoring.



Career Growth
Strengthened skills in SIEM, log analysis, troubleshooting, and incident detection.

“ Every challenge solved is a step closer to mastering cybersecurity. ”

ATTACK SIMULATION WORKFLOW

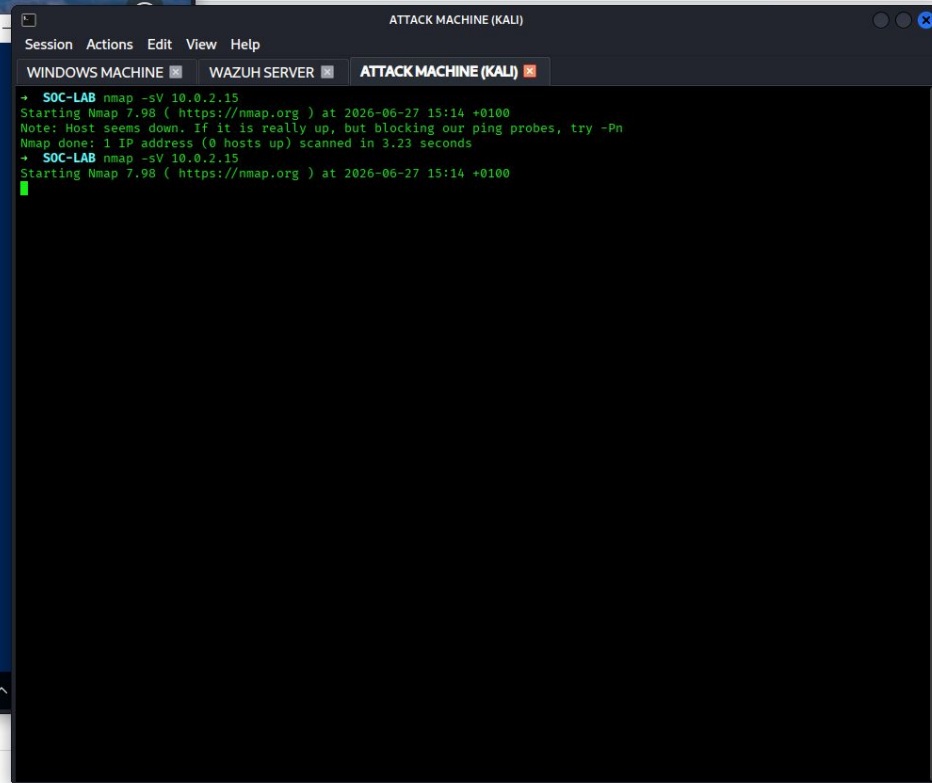
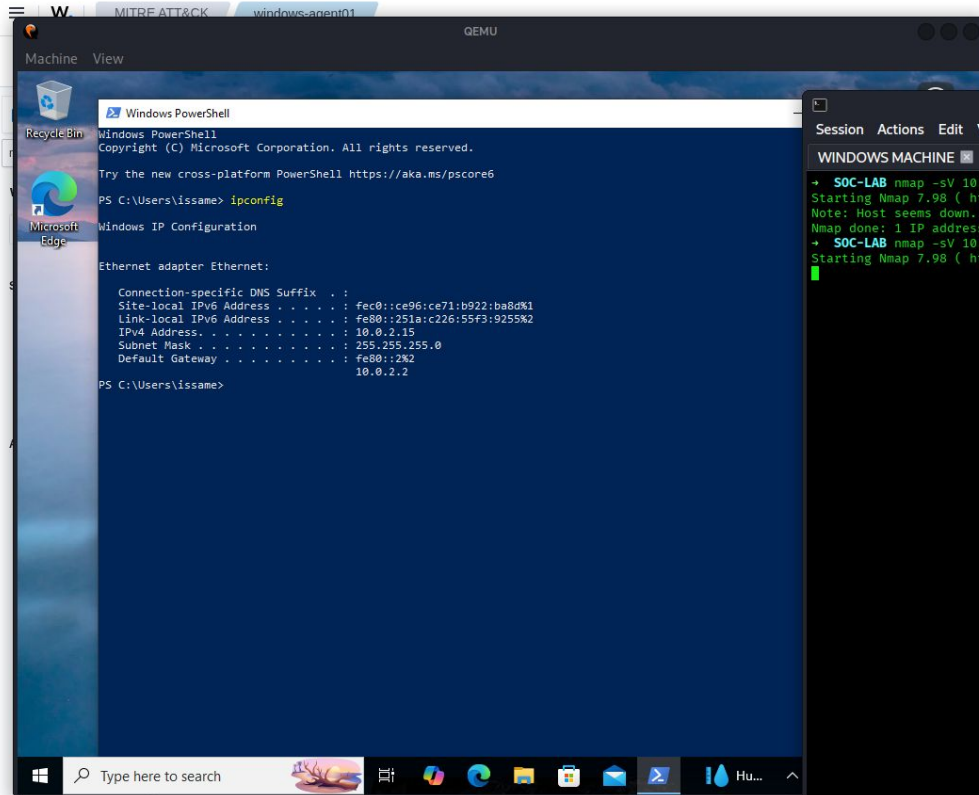
SOC ANALYST VIEW



END-TO-END VISIBILITY. REAL-TIME DETECTION. STRONGER SECURITY.

From attack simulation to alerting – powered by Wazuh.

wazuh.

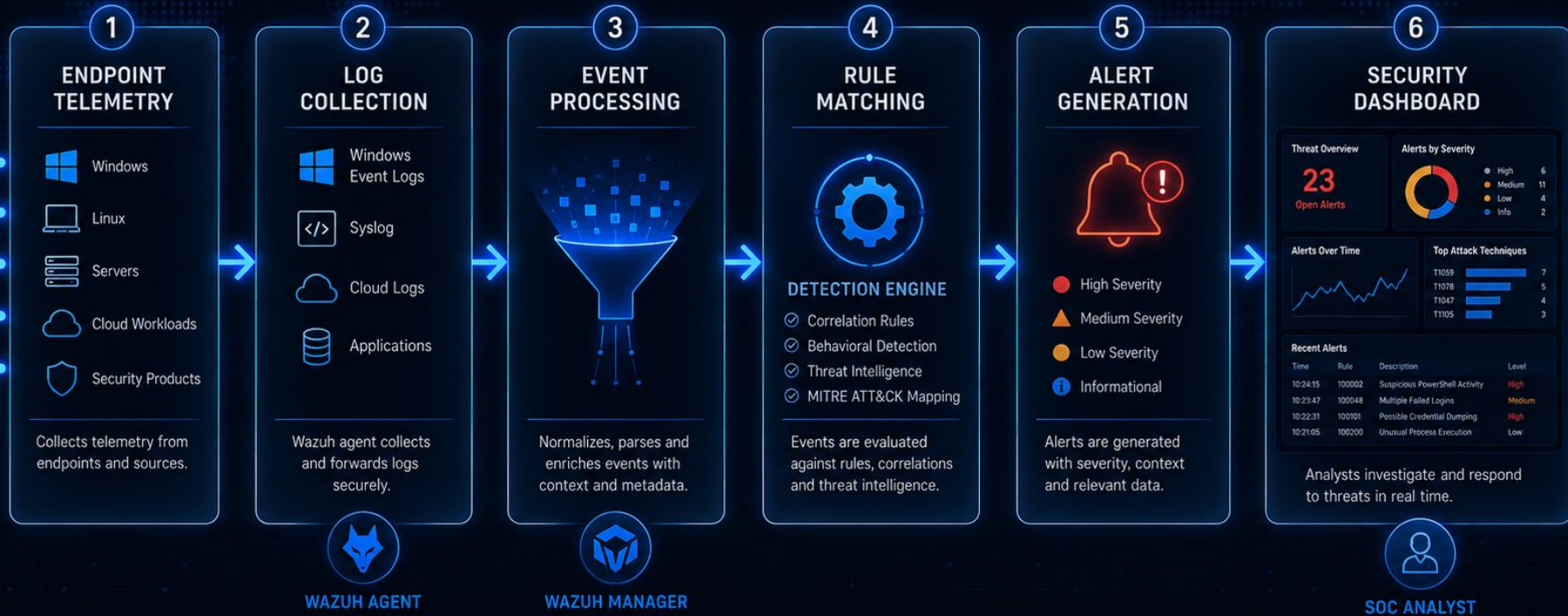


data.win.eventdata.impersonationLevel	>	Jun 27, 2026 @ 15:04:20.241	T1087	Discovery
data.win.eventdata.integrityLevel	>	Jun 27, 2026 @ 15:04:20.209	T1087	Discovery
data.win.eventdata.keyLength	>	Jun 27, 2026 @ 15:04:20.193	T1087	Discovery



SIEM DETECTION PIPELINE

TELEMETRY TO THREAT DETECTION



END-TO-END VISIBILITY
Complete visibility across your environment.



REAL-TIME DETECTION
Detect threats as they happen with low latency.



THREAT INTELLIGENCE
Leverage global threat intel and MITRE ATT&CK.



FASTER RESPONSE
Empower SOC teams to respond with speed and confidence.

wazuh.

SKILLS DEMONSTRATED

Core cybersecurity skills applied in real-world security operations



SIEM MONITORING

Monitor and analyze security events in real-time using SIEM platforms.



LOG ANALYSIS

Collect, parse and correlate logs to identify meaningful insights.



INCIDENT DETECTION

Detect suspicious activities and potential security incidents proactively.



LINUX ADMINISTRATION

Manage and secure Linux systems and services in enterprise environments.



WINDOWS SECURITY

Hardening, monitoring and maintaining Windows systems securely.



THREAT HUNTING

Proactively hunt threats using hypotheses, TTPs and threat intelligence.



NETWORK SECURITY

Monitor and defend networks using traffic analysis and security controls.



VIRTUALIZATION

Deploy, manage and secure virtualized environments and workloads.



ENDPOINT MONITORING

Monitor endpoint health, detect threats and ensure endpoint protection.



SECURE
Protect systems and data



DETECT
Identify threats early



ANALYZE
Investigate and correlate



RESPOND
Take action and mitigate



IMPROVE
Continuously enhance security posture



LESSONS LEARNED



KEY TAKEAWAYS FROM MY SOC LAB PROJECT



01



Built an end-to-end SOC lab

Designed and deployed a complete SOC environment from scratch.

02



Integrated Windows telemetry with Wazuh

Successfully connected Windows event logs and telemetry to Wazuh.

03



Resolved agent version compatibility issues

Troubleshoot version mismatches and ensured stable agent communication.

04



Managed Linux-based SIEM infrastructure

Installed, configured and maintained Wazuh Manager on Linux.

05



Investigated security events and alerts

Analyzed alerts, correlated events and identified potential security threats.

06



Improved incident detection workflows

Built repeatable detection processes and improved alert response efficiency.



STRONGER SECURITY. BETTER VISIBILITY. FASTER RESPONSE.

From telemetry to intelligence – building a SOC you can trust.

wazuh.



END-TO-END SOC MONITORING OPERATIONAL

Real-time Threat Detection. Complete Visibility. Active Protection.



SYSTEM STATUS
ALL SYSTEMS OPERATIONAL



KALI LINUX

Attack Simulation Host

- Tools Executed
- Payload Delivered
- Activities Logged

OPERATIONAL

```
kali@kali:~$  
$ nmap -sS 192.168.1.10  
$ crackmapexec smb 192.168.1.10  
$ powershell -enc SQBF...  
$ mimikatz.exe privilege::debug  
...  
Operation completed successfully
```



WINDOWS 10 ENTERPRISE (CIS BENCHMARK)

Monitored Endpoint

- Events Collected
- Logs Forwarded
- Agent Connected



ENDPOINT HEALTH
HEALTHY
CIS Benchmark
Compliant



WAZUH SIEM

Centralized Monitoring

- Logs Ingested
- Rules Applied
- Alerts Generated

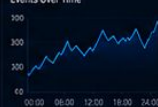
wazuh.

Dashboard

Security Overview

Total Events	Alerts	Active Endpoints	Threats Detected
128,642	247	28	57
↑ 18.6%	↑ 22.7%	↑ 12.0%	↑ 19.0%

Events Over Time



Alerts by Severity



LATEST ALERT

Threat Detected Successfully

Alert Name: Suspicious PowerShell Activity
Rule ID: 100021
MITRE ATT&CK: T1059.001
Severity: High
Status: RESOLVED

LIVE ALERT FEED

- HIGH** Suspicious PowerShell Activity Rule ID: 100021 | T1059.001 Just now
- HIGH** Multiple Failed Logon Attempts Rule ID: 100051 | T1110 1 min ago
- HIGH** Possible Credential Dumping Rule ID: 100036 | T1003 2 min ago

MONITORING METRICS (Real-time)



128,642

Events Processed
↑ 18.6%



247

Alerts Generated
↑ 22.7%



125

Detection Rules Active
↑ 8.7%



28

Endpoints Monitored
All Healthy



57

Threats Detected
↑ 19.0%



2.3s

Mean Time to Detect
↓ 23.4%

ENDPOINT HEALTH



28

Total Endpoints



28

Healthy



0

Issues



ATTACK SIMULATION
Completed Successfully



WINDOWS EVENT LOGS
Collected & Forwarded



WAZUH DETECTION
Triggered & Correlated



SOC VISIBILITY
Achieved End-to-End



MISSION STATUS
SUCCESS

End-to-End SOC Monitoring is Fully Operational